

```

Program Question_1
  Implicit none
  Real, Dimension(5)::x,y
  Real::xp,yp,L
  Integer::i,j
  x=[1,2,4,5,7]
  y=[52,5,-5,-40,10]
  xp=3.0
  yp=0.0
  Do i=1,5
    L=1.0
    Do j=1,5
      if(j/=i)then
        L=L*(xp-x(j))/(x(i)-x(j))
      end if
    End Do
    yp=yp+y(i)*L
  End Do
  Write(*,*)"The interpolated value",yp

End Program

```